

SEDAR 34: HMS Atlantic Sharpnose and Bonnethead Sharks

Atlantic Sharpnose

Data Recommendations

- More research is necessary on review/improvement/development of shrimp bycatch estimation models for both data-poor and data-rich species
- More research is necessary on integration of various local abundance indices into a global abundance index based on spatio-temporal, physical-biological characteristics and variability.

Reviewer Recommendations

Apostolaki Recommendations

The research recommendations that the Panel made are appropriate and will add value to future stock assessments. I have listed here those I consider would make the greatest difference.

- a) Clearly, a priority is to undertake work to provide all the relevant information to run single stock assessments instead of running assessment models for stocks combined.
- b) Good estimates of by-catch in shrimp fisheries are very important since they drive total catch values and thus influence model predictions on MSY. Therefore, work to increase precision in those estimates would improve future model predictions. This refers to both catches before and after 1972.
- c) Identification of additional information/factors that could improve the explanatory power of the standardization models for CPUEs or help understand the contradictory trends among the CPUEs series should also be supported.
- d) Data collection to support calculations of gear selectivity and improve the quality of the results will also be beneficial.
- e) Estimates of post release mortality are based on a small sample size so collection of additional data is recommended as well as species-specific collection of such data.

The SEDAR process is generally effective and achieves its objectives. Based on this specific SEDAR event and report, and previous experience in undertaking desk based reviews, I believe that if reviewers were given the option to have a short meeting (teleconference) with the relevant analysts, that would add value to the process.

Cook Recommendations

Two research recommendations are made in the report which relate to improving the shrimp by-catch estimates and the integration of indices. I would strongly support both of these recommendations. In the case of the shrimp bycatch it is of particular importance to develop a method that is able to reflect the response of the catch to changes in stock abundance. The method used for this assessment uses only effort and is a severe weakness which is bound to result in bias in the estimates.

Assessment models are often a matter of taste where practitioners adopt the model that they prefer. I felt the model used in this assessment demanded far too much from the data and as a consequence a very large number of assumptions had to be made which can only be examined by a very complex set of

sensitivity tests whose results themselves are not easy to digest or interpret. I would recommend that an assessment model that is more closely designed around the available data is developed. If, in reality, shrimp bycatch data cannot be estimated with any precision, then it might be better to treat these catches as unknown and use the effort data directly in the model. It would also be preferable to describe the population in terms of length rather than age.

The assessment report documents are very well written and presented. They were available on time and all this made the work of review much easier. I missed the opportunity to raise questions about the assessments with the assessment panel. It might be useful to try to organize a webinar at some stage during the review so that reviewers can seek clarification from the assessment panel to avoid mistakes and misunderstandings in their reports. I imagine that some of my comments may have arisen through incomplete understanding of what was done.

Rice Recommendations

There are sufficient data for further development of these assessments using sex-structured models that estimate selectivity inside an assessment model, fitting to size data, not length data converted to age. The assessment team mentioned that this is a desirable framework for the next assessment. Specific recommendations for research priorities include:

- To develop region specific indices of abundance. I note that this had been done, alongside the combined indices.
- Ensuring information about sex, length and age is collected in each fishery, throughout with respect to the spatial/temporal nature of the fishery.
- Research that improves the understanding of historical landings in the shrimp fishery, both in the modern and historical period and to support the assumptions about when stocks are at virgin biomass if this assumption is carried forward in future assessments.
- Research into the change in selectivity resulting from regulatory or gear changes (e.g. TEDs).

With respect to the overall SEDAR process, it is apparent that the some aspect of the process failed. The major issue detracting from the quality of any advice based on the bonnethead shark stock assessment is that it does not represent the status of either of the regional stocks.

Bonnethead

Data Recommendations

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